CleaveNet: designing protease substrates

Deep learning guided design of protease substrates

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The design goal is a cleavage profile

Protease-activated probes and drugs need a peptide trigger that is cleaved in the intended biological setting.

Efficient
Strong cleavage by the target enzyme.

A strong MMP13 substrate may also be a strong substrate for other MMPs.

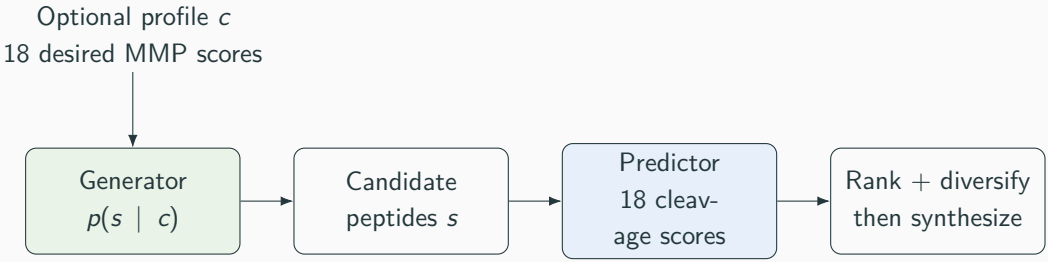
Selective
Preferential cleavage by the target over competing enzymes.

Perfect discrimination can come with a weak absolute signal.

Question: can sequence models design new peptides with both properties?

Introduction; Fig. 1. MMP13 is a collagenase relevant to osteoarthritis, tissue remodeling and cancer.

Two learned models replace nomination and virtual screening



- The generator proposes sequences; the predictor estimates their properties.
- Conditioning changes where the generator samples; ranking remains necessary.
- The final evidence is cleavage of synthesized peptides by recombinant MMPs.

Fig. 1; schematic redrawn for these notes. c is a score profile, not a target protein sequence.

What data define the sequence–function landscape?

Source	Sequences and readout	Role
mRNA display	18,583 synthetic 10-mers; normalized Z-scores across 18 MMPs	Training and within-assay evaluation
Prior fluorescence panel	71 peptides, 7–14 residues; screened against 7 MMPs	Independent assay and sequence-distribution test
New fluorogenic screen	95 designed/control peptides $\times$ 12 MMPs	Prospective design validation

A training Z-score is a relative assay readout, not k_{cat}/K_M or a universal cleavage probability.

Methods: Datasets; Fig. 2A; Fig. 7A. Training data: Kukreja et al., Chem. Biol. (2015).

Keep three different uses of “test” separate

1. Split 18,583 sequences 80/20: 14,866 training and 3,717 initial test peptides.
2. Remove 816 test peptides with minimum Levenshtein distance < 3 from training: 2,901 filtered test peptides remain.
3. Within the original training portion, train five predictors on independent 80/20 training/validation splits; use their spread as uncertainty.

The 71-peptide fluorescence panel tests transfer across assay and sequence composition. It is a stronger distribution shift than the display split.

Evaluation caveat
Methods also describe hyperparameter and generator checkpoint selection by test loss. The test set is therefore not a pristine model-selection holdout.

Methods: Datasets; Training the CleaveNet predictors/generator. Some generation-distribution plots use the unfiltered 3,717 peptides.

Predictor: sequence in, 18 continuous scores out

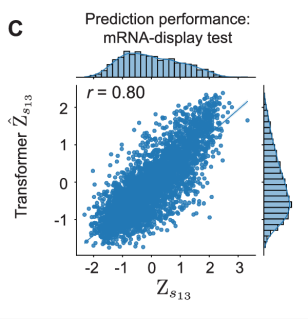
$$s \longrightarrow \text{amino-acid embeddings} + \text{positions} \longrightarrow \text{encoder} \longrightarrow \widehat{\boldsymbol{Z}}_s \in \mathbb{R}^{18}$$

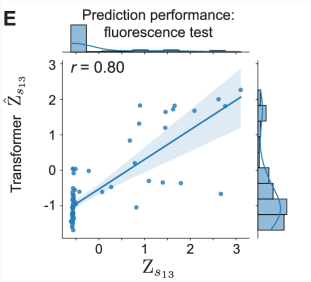
- Transformer: two encoder layers, approximately 56k parameters; a [CLS] representation pools the sequence for multi-output regression.
- A bidirectional LSTM is a competitive alternative; the transformer is used for subsequent pipeline analyses.
- Five trained predictors produce a mean $\widehat{Z}_{sm}$ and standard deviation σ_{sm} for peptide s and MMP m.

Practical interpretation
A high predicted score supports ranking a candidate above others in this learned landscape. Ensemble disagreement flags some uncertain rankings; it is not a calibrated confidence interval.

Fig. 2A–B; Methods: Training the CleaveNet predictors; Supplementary Fig. 10.

Prediction transfers, but absolute scores remain assay dependent





MMP13 correlation:

- Display test: $r = 0.80$, $n = 2,901$.
- Fluorescence test: $r = 0.80$, $n = 71$.

Transformer MAE for MMP13: 0.468 ± 0.004 versus 0.610 ± 0.055 in the two assays.

For assay transfer, ranking is more interpretable than equality of predicted and observed Z.

Fig. 2C,E (cropped; axes retained); Supplementary Table 1. Author figures, CC BY 4.0.

Generator: learn joint sequence patterns, then sample

Introduce peptide residues $x_1, \ldots, x_L$ and optional desired profile c:

$$p(s \mid c) = \prod_{i=1}^{L+1} p(x_i \mid x_1, \ldots, x_{i-1}, c), \qquad x_{L+1} = \text{[STOP]}.$$

- Decoder-only transformer: three layers, approximately 328k parameters.
- Train by next-token prediction. Half the examples use [START]; half use the 18-score conditioning vector, rounded to one decimal place.
- Sample each residue from its predicted distribution until [STOP]. Sequence length can vary, but comparison analyses retain 10-mers.

This is a small task-specific model learned from display data, not a pretrained general protein language model.

Methods: Training the CleaveNet generator; unconditional generations; Supplementary Table 2. Equation adapted with conditioning explicit.

A concrete conditioning example

Toy input: three of the 18 entries

Profile	MMP13	MMP2	MMP9
Efficient	3.0	2.6	2.4
Selective	2.5	0.1	0.2

The efficient profile tolerates off-target cleavage; the selective profile asks the model to avoid it.

Autoregressive operation

$c \rightarrow$ first-residue distribution

$c, \mathtt{F} \rightarrow$ next-residue distribution

$c, \mathtt{FA} \rightarrow$ next-residue distribution

$\ldots \rightarrow$ [STOP]

No single sequence is determined by c. Multiple samples explore sequences compatible with the same requested behavior.

Illustrative numbers and partial prefix, not model outputs. Actual MMP13 experiment: top-50 seed profiles $\times$ 400 samples = 20,000 candidates. Methods: Conditional generations.

Efficiency and selectivity rank different candidates

For M modeled MMPs, the design-time selectivity score is

$$\widehat{S}_{sm} = \widehat{Z}_{sm} - \frac{1}{M-1}\sum_{i \neq m} \widehat{Z}_{si}.$$

Toy peptide	$\widehat{Z}_{13}$	Mean other $\widehat{Z}$	σ_{13}	$\widehat{Z}_{13} - \sigma_{13}$	$\widehat{S}_{13}$
A	3.0	2.5	0.2	2.8	0.5
B	2.2	0.1	0.1	2.1	2.1

- Efficiency nomination prefers A; selectivity nomination prefers B.
- An average-off-target penalty can hide one problematic competing enzyme. Inspect the entire predicted profile before choosing a probe.

Methods Eq. 3 and candidate selection. Toy values invented only to explain the score. Experimental selectivity values later use the authors' assay metric.

Why compare with site-independent sequence sampling?

Site-independent baseline
Estimate amino-acid frequencies separately at each peptide position, then sample each position independently.

It preserves marginal composition but loses dependencies between residues.

CleaveNet
Predict the next residue using the full prefix and, optionally, the target score profile.

It can retain multi-residue motifs associated with cleavage, such as the frequent MMP13 trimer `PLG`.

The stronger experimental baseline adds the same predictor to site-independent proposals. This separates the value of learned screening from the value of learned generation.

Figs. 3A, 4B, 5B; Methods: Selection of baseline and control substrates. Motif enrichment supports learned dependencies, not a causal mechanism.

Unconditional sampling reproduces useful sequence structure

- The generator produced 20,000 candidates; 95 exact matches were removed, yielding 19,905 sequences for reported generation metrics.
- Generated sequences recover MMP-like motifs and similar computed biophysical distributions. Around the cleavage region P3–P2′, mean position-wise KL divergence is 0.25 versus 0.404 for site-independent sampling.
- Predicted cleavage profiles and recurring k-mers resemble display peptides more than the independent baseline does.

What this establishes
The model learns a plausible sequence distribution with limited exact copying. Predictor-scored plausibility alone cannot establish experimental function or broad extrapolation.

Figs. 3–4; Supplementary Figs. 13–15. Methods separately report filtering 87 non-10-mers; the sampling/filter accounting is not fully explicit.

Conditional sampling enriches a desired region of the landscape

1. Predict all display peptides' profiles and identify the top 50 for MMP13 selectivity.
2. Use each rounded 18-score profile to generate 400 peptides: 20,000 proposals in total.
3. Re-score the proposals with the predictor and compare against unconditional proposals and both site-independent baselines.

Reported median predicted MMP13 selectivity is 5.5-fold higher for conditional than unconditional CleaveNet proposals ($p < 10^{-4}$). A similar MMP9 experiment supports in-silico portability across targets; prospective selective design is experimentally tested here for MMP13.

Fig. 6B–D; Supplementary Fig. 17 and Table 5. Score enrichment is measured with the same predictor used for selection.

Candidate selection differs between the two design tasks

Step	Efficient MMP13 designs	Selective MMP13 designs
Proposal pool	20,000 unconditional samples	20,000 samples conditioned on selective profiles
Ranking	$\widehat{Z}_{13} - \sigma_{13}$	$\widehat{S}_{13}$; no uncertainty filter
Diversity	Retain top-scoring candidates without overlapping 5-mers	Same 5-mer redundancy filter
Synthesis	Top 24 peptides	Top 24 peptides

The tested hit rate describes selected candidates after virtual screening, not random generator outputs.

Methods: Selection of CleaveNet-generated substrates for in vitro experiments; Figs. 5A and 7A.

Prospective screen: 95 peptides, not 95 generated designs

Source	Efficient	Selective
Generator + predictor	24	24
Independent + predictor	8	8
Site-independent alone	8	8
Display positive controls	5	5

Add 5 display negative controls:
$45 + 45 + 5 = 95$ substrates.

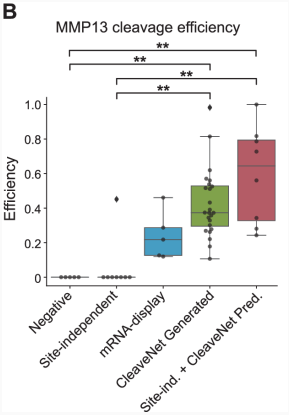
Assay
FRET fluorogenic peptides against 12 recombinant MMPs.

Two independent runs, each with duplicate reactions per protease–substrate pair.

$95 \times 12 = 1,140$ distinct pairs, before replication.

Fig. 7A; Supplementary Fig. 18; in vitro Methods. Use these consistent figure counts; Methods baseline-selection prose says five rather than eight.

Efficiency: learned screening is already very powerful



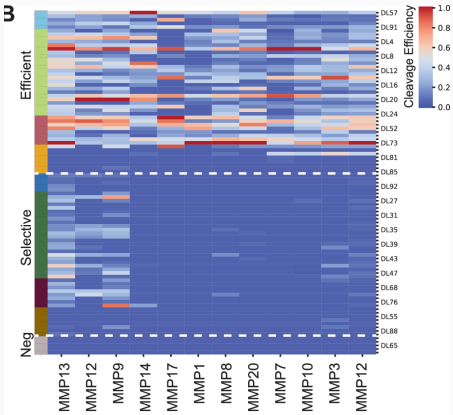
Nomination group	Cleaved	Median E
Site-independent	1/8	0.00
Independent + predictor	8/8	0.64
Generator + predictor	24/24	0.37
Display efficient controls	5/5*	0.22

The predictor rescues the baseline hit rate. The strongest median efficiency is achieved by predictor-ranked independent proposals.

CleaveNet's generator is therefore not shown to dominate every useful proposal strategy.

Fig. 5B; Supplementary Table 3. E is normalized assay efficiency. *Main-text/figure control count; Supplementary Table 3 instead labels 4/4. Figure cropped, CC BY 4.0.

Cross-protease measurements reveal the intended shift



Each row is a substrate; each column an assayed MMP. Colors show normalized cleavage efficiency.

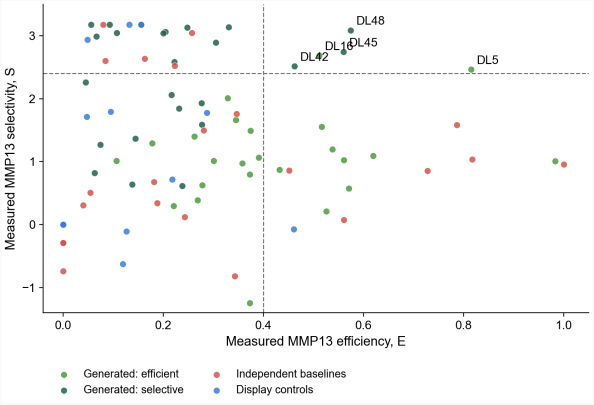
Efficient designs (upper block) often show cleavage across several MMPs.

Selective designs (middle block) concentrate cleavage toward MMP13.

Both 24-peptide CleaveNet groups are cleaved by MMP13: 48/48 selected designs. This does not imply all 48 are uniquely selective.

Fig. 7B (cropped, CC BY 4.0); Supplementary Figs. 18, 24; Supplementary Table 3.

The useful advance is access to a better tradeoff



Across all 95 peptides, authors set $E > 0.4$ and $S > 2.4$ as high thresholds.

Quadrant	n
Low E, low S	56
Low E, high S	19
High E, low S	15
High E, high S	5

All five upper-right hits were fully CleaveNet-designed: two from efficient nomination and three from selective nomination.

Fig. 7D redrawn from Source Data, all 95 points; baseline/control subgroups pooled in legend. Supplementary Table 8. These are authors' experimental metrics, not the toy predicted scores.

Concrete hits: selective and efficient are distinct achievements

Peptide	Sequence	Experimental E	Experimental S
DL48	FAPRQMFDPG	0.57	3.09
DL45	VAPYGFRTAS	0.56	2.74
DL16	PVPLGVRDTA	0.51	2.69
DL42	TIPYGFWGPY	0.46	2.52
DL5	TAPLGIRDGT	0.81	2.46

These five hits occupy the high-efficiency/high-selectivity quadrant.

A separate set—DL41, DL32 and DL28—was uniquely cleaved by MMP13 in the 12-MMP panel, with lower efficiency. Selective display controls also reached this maximal selectivity, at $E < 0.16$.

Fig. 7D–E; Supplementary Tables 7–8. “Uniquely” is relative to the assayed MMP panel and assay conditions.

Motifs are hypotheses about recognition, not proof of mechanism

Using the conventional cleavage-site notation:

$$\cdots \text{P4–P3–P2–P1} \downarrow \text{P1}'\text{–P2}'\text{–P3}' \cdots$$

- Efficient MMP13 substrates: top-seven measured hits share PL at P3–P2, suggesting a preference for leucine at P2.
- High-selectivity/low-efficiency substrates enrich arginine at P2, aromatic residues at P1′ and aspartate at P3′.
- High-efficiency/high-selectivity sequences combine traits of both classes; this is a small, selected group of five peptides.

Follow-up: mutate P2 and P1′ in matched backgrounds and measure kinetics across MMPs.

Figs. 5C and 7F; Supplementary Fig. 16. Follow-up is our proposal; motifs inferred from sequence alignments do not directly map every cleavage bond.

Calibration connects display scores to measurable cleavage

The new screen yields MMP-specific predicted-score thresholds, spanning 0.3–2.4 (MMP13: 1.0).

$$\widehat{E}_{sm} = \max\left(0, \frac{\widehat{Z}_{sm} - T_m}{\max_s(\widehat{Z}_{sm} - T_m)}\right)$$

- Subtract the learned cleavage threshold T_m, set subthreshold values to zero, and scale above-threshold scores within the evaluated set.
- This yields a corrected predicted efficiency, not a measured kinetic constant. Its scale depends on which sequences define the maximum.
- Six unassayed MMPs receive an imputed average threshold of 0.983; their calibration is less direct.

Threshold fitting and classifier evaluation on the same 95-peptide panel need fresh validation.

Methods Eqs. 4–5; Supplementary Table 6 and Fig. 25. Equation rewritten for positive-denominator case; T_m is estimated from fluorogenic screening.

What remains uncertain?

1. **Scope.** Short synthetic peptides and MMPs; no demonstration of tissue selectivity, therapeutic activation or generalization to other protease classes.
2. **Assay comparability.** Normalized fluorescence readouts are not enzyme kinetics; enzyme concentrations, incubation times and buffers differ across MMPs.
3. **Selection and circularity.** Predicted enrichment reuses the selection model; prospective cleavage is persuasive, but only 48 selected generator designs are tested.
4. **Evaluation independence.** Test-loss model selection, assay-derived threshold tuning, and strong overlap in short sequence motifs temper extrapolation claims.

Discussion; Methods; Supplementary Tables 4, 6, 8. These are our evaluation caveats, distinguished from the authors' demonstrated results.

Journal club discussion: what would change our confidence?

Most convincing result
Prospective cleavage profiles include five efficient/selective hits, with appropriate proposal and screening baselines. The models help search useful combinations beyond the tested display controls.

1. How much does conditioning add at a fixed synthesis budget? Compare conditional and unconditional proposals with identical ranking/diversity rules.
2. Can a fresh library confirm the cleavage thresholds and the high-efficiency/high-selectivity region?
3. Do selective hits retain useful rates in biological fluids, among other protease classes, and in an actual activated probe or drug?

Our interpretation and proposed next experiments, based on Figs. 5–7. Current evidence supports a practical design pipeline, not clinical specificity.

Sources and reusable resources

Journal version

Martin-Alonso C., Alamdari S., Samad T.S. et al. *Deep learning guided design of protease substrates*. Nature Communications **17**, 54 (2026). Published 6 January 2026. doi.org/10.1038/s41467-025-67226-1

Accessed material

Full journal article including Methods; Supplementary Information, 28 pages (25 figures and 9 tables); Source Data Fig. 7D (95 rows). Selected panels cropped; tradeoff plot redrawn with pooled legend groups.

Code, data and model weights

github.com/microsoft/cleavenet

Original figure material: authors, CC BY 4.0. Explanatory schematics and toy calculations are adaptations for these notes; toy sequences/scores are not experimental predictions.

Source access: 1 November 2026. See README.md for source-count inconsistencies and validation scope.